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1) write a c program for sum of two matrices

code:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int rows, cols, i, j;

    printf("Enter the number of rows: ");

    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("\nEnter elements of First Matrix:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("\nEnter elements of Second Matrix:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }
    for(i = 0; i < rows; i++)
```

```
    {
        for(j = 0; j < cols; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }

    // Display result
    printf("\nSum of the two matrices:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            printf("%d\t", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
Enter the number of rows: 2
Enter the number of columns: 2

Enter elements of First Matrix:
1 2
3 4

Enter elements of Second Matrix:
5 6
7 8

Sum of the two matrices:
```

6 8
10 12

2) write a c program to read n number of values in an array and display them in reverse order

code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Array elements in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

Output:

```
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
```

Array elements in reverse order:
50 40 30 20 10

3) write a c program to count the total numbers of duplicate in an array

code:

```
#include <stdio.h>
int main()
{
    int arr[100], n;
    int i, j, count = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(arr[i] == arr[j])
```

```
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d\n", count);

    return 0;
}
```

Output:

```
Enter the number of elements: 7
Enter 7 elements:
1 2 3 2 4 5 1
Total number of duplicate elements = 2
```

4) write a c program to print unique element in an array

code:

```
#include <stdio.h>
int main()
{
    int arr[100], n;
    int i, j, count;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
```

```
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Unique elements in the array are:\n");
    for(i = 0; i < n; i++)
    {
        count = 0;
        for(j = 0; j < n; j++)
        {
            if(arr[i] == arr[j])
            {
                count++;
            }
        }

        if(count == 1)
        {
            printf("%d ", arr[i]);
        }
    }
    printf("\n");
    return 0;
}
```

Output:

Enter the number of elements: 8

Enter 8 elements:

1 2 3 2 4 5 1 6

Unique elements in the array are:

3 4 5 6

5) write a c program to insert in sorted array

code:

```
#include <stdio.h>

int main()
{
    int arr[100], n, element;
    int i, pos;

    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements in ascending order:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Enter the element to insert: ");
    scanf("%d", &element);
    pos = n;
    for(i = 0; i < n; i++)
    {
        if(arr[i] > element)
        {
            pos = i;
```

```
            break;
        }
    }

    for(i = n; i > pos; i--)
    {
        arr[i] = arr[i - 1];
    }
    arr[pos] = element;
    n++;

    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }

    printf("\n");

    return 0;
}
```

Output:

```
Enter the number of elements: 5
Enter 5 elements in ascending order:
10 20 30 40 50
Enter the element to insert: 35

Array after insertion:
10 20 30 35 40 50
```

6) write a c program to insert in unsorted array

code:

```
#include <stdio.h>

int main()
```

```
{
    int arr[100], n;
    int element, position, i;

    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }


    printf("Enter the element to insert: ");
    scanf("%d", &element);

    printf("Enter the position (1 to %d): ", n + 1);
    scanf("%d", &position);


    if(position < 1 || position > n + 1)
    {
        printf("Invalid position!\n");
    }
    else
    {
        for(i = n; i >= position; i--)
        {
            arr[i] = arr[i - 1];
        }

        arr[position - 1] = element;
        n++;

        printf("Array after insertion:\n");
```

```
        for(i = 0; i < n; i++)
        {
            printf("%d ", arr[i]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:
Enter the number of elements: 5
Enter 5 elements:
10 30 20 50 40
Enter the element to insert: 25
Enter the position (1 to 6): 3

Array after insertion:
10 30 25 20 50 40

7) write a c program to matrix multiplication
code:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2;
    int i, j, k;

    printf("Enter rows and columns of First Matrix: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of Second Matrix: ");
    scanf("%d%d", &r2, &c2);
```

```
if(c1 != r2)
{
   printf("Matrix multiplication is not possible.\n");
   return 0;
}

printf("Enter elements of First Matrix:\n");
for(i = 0; i < r1; i++)
{
   for(j = 0; j < c1; j++)
   {
      scanf("%d", &a[i][j]);
   }
}

printf("Enter elements of Second Matrix:\n");
for(i = 0; i < r2; i++)
{
   for(j = 0; j < c2; j++)
   {
      scanf("%d", &b[i][j]);
   }
}


for(i = 0; i < r1; i++)
{
   for(j = 0; j < c2; j++)
   {
      c[i][j] = 0;
   }
}
for(i = 0; i < r1; i++)
{
```

```
        for(j = 0; j < c2; j++)
        {
            for(k = 0; k < c1; k++)
            {
                c[i][j] = c[i][j] + a[i][k] * b[k][j];
            }
        }
    }
    printf("\nProduct of the two matrices:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            printf("%d\t", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
Enter rows and columns of First Matrix: 2 2
Enter rows and columns of Second Matrix: 2 2

Enter elements of First Matrix:
1 2
3 4

Enter elements of Second Matrix:
5 6
7 8

Product of the two matrices:
19      22
43      50
```

8) write a c program to matrix transpose

code:

```
#include <stdio.h>

int main()
{
    int a[10][10], transpose[10][10];
    int rows, cols, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    printf("Enter the number of columns: ");
    scanf("%d", &cols);

    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < rows; i++)
    {
        for(j = 0; j < cols; j++)
        {
            transpose[j][i] = a[i][j];
        }
    }

    printf("\nTranspose of the matrix:\n");
    for(i = 0; i < cols; i++)
```

```
    {
        for(j = 0; j < rows; j++)
        {
            printf("%d\t", transpose[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output:

```
Enter the number of rows: 2
Enter the number of columns: 3

Enter the elements of the matrix:
1 2 3
4 5 6

Transpose of the matrix:
1   4
2   5
3   6
```

9) write a c program to right diagonal sum

code:

```
#include <stdio.h>

int main()
{
    int a[10][10];
    int n, i, j, sum = 0;

    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);
```

```
    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i + j == n - 1)
            {
                sum = sum + a[i][j];
            }
        }
    }

    printf("Sum of the right diagonal = %d\n", sum);

    return 0;
}
```

Output:

```
Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 2 3
4 5 6
7 8 9

Sum of the right diagonal = 15
```

10) write a c program to check for identity matrix

code:

```c
#include <stdio.h>

int main()
{
    int a[10][10];
    int n, i, j, flag = 1;

    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);

    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j)
            {
                if(a[i][j] != 1)
                {
                    flag = 0;
                    break;
                }
            }
            else
            {
                if(a[i][j] != 0)
                {
```

```
                flag = 0;
                break;
            }
        }
    }

    if(flag == 0)
    {
        break;
    }
}

if(flag == 1)
{
    printf("The matrix is an Identity Matrix.\n");
}
else
{
    printf("The matrix is NOT an Identity Matrix.\n");
}

return 0;
}
```

Output 1:

```
Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 0 0
0 1 0
0 0 1
The matrix is an Identity Matrix.
```

Output 2:

```
Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 2 0
```

0 1 0
0 0 1

The matrix is NOT an Identity Matrix.

11) write a c program to find no vowels ,consonants, digits and white spaces

code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;
    int vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);


    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i]
<= 'z'))
        {
            if(str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O'
|| str[i] == 'U' ||
               str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o'
|| str[i] == 'u')
            {
                vowels++;
            }
            else
            {
                consonants++;
```

```
            }
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(str[i] == ' ')
        {
            spaces++;
        }
    }


    printf("Number of vowels      = %d\n", vowels);
    printf("Number of consonants  = %d\n", consonants);
    printf("Number of digits      = %d\n", digits);
    printf("Number of white spaces= %d\n", spaces);

    return 0;
}
```

Output:

```
Enter a string: Hello World 123

Number of vowels      = 3
Number of consonants  = 7
Number of digits      = 3
Number of white spaces= 2
```

12) write a c program to frequency of a character in a string

code:

```
#include <stdio.h>

int main()
{
```

```
    char str[100], ch;
    int i, count = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter the character to find: ");
    scanf("%c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
        {
            count++;
        }
    }

    printf("Frequency of '%c' = %d\n", ch, count);

    return 0;
}
```

Output:

```
Enter a string: programming
Enter the character to find: g
Frequency of 'g' = 2
```

13) write c program to count the words in a string

code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, words = 0;

    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((i == 0 && str[i] != ' ' && str[i] != '\n') ||
           (str[i] != ' ' && str[i - 1] == ' '))
        {
            words++;
        }
    }

    printf("Number of words = %d\n", words);

    return 0;
}
```

Output:

```
Enter a string: C programming is easy
Number of words = 4
```

14) write a c program to sort a string an array

code:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][100], temp[100];
    int n, i, j;

    printf("Enter the number of strings: ");
    scanf("%d", &n);
    getchar();

    printf("Enter %d strings:\n", n);
    for(i = 0; i < n; i++)
```

```
    {
        fgets(str[i], sizeof(str[i]), stdin);
        str[i][strcspn(str[i], "\n")] = '\0';
    }

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("\nStrings in alphabetical order:\n");
    for(i = 0; i < n; i++)
    {
        printf("%s\n", str[i]);
    }

    return 0;
}
```

Output:

```
Enter the number of strings: 5
Enter 5 strings:
Banana
Apple
Orange
Mango
Grapes
```

Strings in alphabetical order:
Apple
Banana
Grapes
Mango
Orange

15) write a c program to toggle case of sentence

code:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i;

    printf("Enter a sentence: ");

    fgets(str, sizeof(str), stdin);

    // Toggle case
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }

    printf("Sentence after toggling case:\n%s", str);
```

```
    return 0;
}
```

Output:
Enter a sentence: Hello World 123

Sentence after toggling case:
hELLO wORLD 123